

MCQ Masterclass

Lesson 48 — Full-Course Rapid Revision

JKSSB Computer Awareness

October 8, 2026

How to Use This Deck

- ▶ This is a **capstone drill**: one high-yield trap per frame, drawn from every unit of the course.
- ▶ For each frame ask yourself the question **before** reading the answer box.
- ▶ **Gold** marks the fact the examiner will test; **gray** marks the wrong belief the distractor relies on.
- ▶ The trap IDs (TRAP-01, ...) match the course trap registry.

The one idea

Every JKSSB computer item is a near-neighbour contrast: what it is, what it is not, and the nearest confusion.

Memory Volatility: RAM vs ROM

- ▶ **RAM** (Random Access Memory) is **volatile**: contents are lost when power goes off.
- ▶ **ROM** (Read-Only Memory) is **non-volatile**: it keeps its contents without power.
- ▶ In normal operation ROM is read-only; rewriting it needs a special process.

Trap alert (TRAP-01, TRAP-02)

RAM is non-volatile / ROM allows normal read and write — both are swapped pairs.
RAM is volatile; ROM is non-volatile and read-only.

Memory Order and Cache

- ▶ Speed order, fastest first: **registers** → **cache (L1, L2, L3)** → **main memory (RAM)** → **secondary storage**.
- ▶ **Cache is faster** than main memory; it holds recently and frequently used data.
- ▶ Cache holds copies of data that live in RAM; it is **not** a backup.
- ▶ The **CPU** works with main memory; it does **not** access secondary storage directly.

Trap alert (TRAP-03)

CPU reads the hard disk directly — wrong. Secondary storage is reached via main memory and the OS.

Input vs Output Classification

- ▶ **Input**: light pen, punch-card reader, digitizer tablet, scanner, MICR reader.
- ▶ **Output**: plotter, drum printer, sound card.
- ▶ Large engineering drawings → **plotter**; magnetic-ink characters → **MICR**.

Trap alert (TRAP-12, TRAP-25)

Plotter / printer are input; sound card is input — all output. Keep MICR (magnetic ink), OCR (printed text), and OMR (marks) distinct.

Hardware vs Software, and BIOS

- ▶ A **device driver** is **software**; the **operating system** is **system software**.
- ▶ **Open-source** means the licence lets you view, change, and share the source — not merely free of cost.
- ▶ The firmware that initialises hardware and loads the OS is the **BIOS** (Basic Input/Output System).

Trap alert (TRAP-10, TRAP-23, TRAP-24)

Driver is hardware; open-source = freeware; the kernel loads the OS — wrong on all three. Driver = software; BIOS boots the machine.

Operating System Facts

- ▶ The OS is the layer that provides a **user-friendly interface** between user and hardware.
- ▶ While running, the OS is resident in **primary storage** (main memory).
- ▶ **Multitasking** means running more than one program at a time.
- ▶ **Android** is the open-source mobile OS; the control unit controls the **operation sequence and data flow**.

Quick check

OS in primary storage + multitasking + Android-as-open-source: three direct-recall items from one frame.

MS Word: Checks, Merges, Breaks

- ▶ **F7** runs spell check; **Mail Merge** combines letters with an address database.
- ▶ A **Section Break** enables different headers and footers; a Page Break does not.
- ▶ Smallest unit of data: **bit**; 8 bits = 1 byte.

Trap alert (TRAP-13, TRAP-27, TRAP-29)

Page Break changes headers; byte is smallest; mail merge = track changes — all wrong. Section Break, bit, Mail Merge.

MS Excel: The Three Symbols

- ▶ Every formula must start with **=**.
- ▶ **\$A\$1** is an absolute reference: neither column nor row moves when copied.
- ▶ **COUNTIFS** counts rows meeting several criteria; **SUMPRODUCT** with multiplication does the same job.

Trap alert (TRAP-14, TRAP-30)

A formula can start with **+** or **%** — the exam answer is **=**. Master the **\$A\$1** / **COUNTIFS** pair; it recurs every cycle.

Access, PowerPoint, PDF and Extensions

- ▶ The query that retrieves data on a condition is the **Select query**; Update, Append, and Delete modify data.
- ▶ **F5** starts the slide show from the beginning.
- ▶ Say **extension** and **format** separately: .png is an extension; PNG is the format.

Trap alert (TRAP-31, TRAP-32, TRAP-34)

F7 starts the show; any query retrieves — wrong. F5; Select query. Never conflate extension with format.

Internet, WWW and ISP

- ▶ The **Internet** is the global network; the **WWW** (World Wide Web) is the system of linked pages served over it.
- ▶ **ISP** — Internet Service Provider — provides the connection.
- ▶ Decimal 10 = binary **1010**; third generation = **ICs**, fourth = microprocessors.

Trap alert (TRAP-26)

ICs arrived in the fourth generation — wrong. ICs → third; microprocessors → fourth.

URL, DNS and @

- ▶ A **URL** (Uniform Resource Locator) is the address of a resource; **DNS** (Domain Name System) resolves names to IP addresses.
- ▶ **IPv4** = 32-bit; **IPv6** = 128-bit with a fixed 40-byte header.
- ▶ In an email address, **@** separates the **user name** from the **domain name**.

Trap alert (TRAP-18, TRAP-28)

IPv6 is 256-bit; @ separates domain and host — both wrong. 128-bit; user name vs domain name.

Email: BCC, POP3, SSL/TLS

- ▶ **BCC** hides recipients from each other; the sender always knows, and To/Cc stay visible.
- ▶ **POP3** — Post Office Protocol version 3 — **downloads** mail to the local machine; IMAP syncs with the server.
- ▶ Secure transmission uses **SSL/TLS**.

Trap alert (TRAP-16, TRAP-17)

BCC hides everyone from everyone; POP3 keeps mail synced — wrong. BCC hides recipients from each other; POP3 downloads locally.

Networking Devices and Protocols

- ▶ **BGP** is inter-domain (between autonomous systems); OSPF and RIP are intra-domain.
- ▶ **ICMP** is a control/diagnostic protocol — no ports, no application data transfer.
- ▶ **HTTP/3** uses **QUIC over UDP**, not TCP.

Trap alert (TRAP-19, TRAP-20, TRAP-21)

BGP routes a LAN; ICMP carries data; HTTP/3 runs on TCP — all three reversed.
Inter-domain; diagnostic; QUIC over UDP.

Malware, Encryption and Cloud

- ▶ Keep virus, worm, and Trojan distinct: a **virus** needs a host, a **worm** self-propagates, a **Trojan** poses as useful software.
- ▶ **Encryption** gives confidentiality; a **digital signature** gives authenticity and integrity.
- ▶ In **IaaS**, the provider manages infrastructure (networking and storage hardware); the customer manages OS, runtime, apps, and data.

Trap alert (TRAP-15, TRAP-22)

Signature encrypts the body; IaaS provider manages the OS — wrong. Signature $\neq$ encryption; provider manages hardware only.

IT Act and E-Governance

- ▶ The **Information Technology Act, 2000** is India's primary cyber law; **Section 43** covers unauthorised access and data damage, **Section 66** covers computer-related offences.
- ▶ **G2C** = Government to Citizen; **G2B** = Government to Business; **G2G** = Government to Government; **G2E** = Government to Employees.
- ▶ Anchor services: **Digital India**, **NeGP/e-Kranti**, and the **NeGD** (National e-Governance Division).

Keep the pair straight

Section 43 = unauthorised access; Section 66 = computer offences. G2C serves the citizen; G2B serves business.

Lesson 48: Recall

- ▶ RAM volatile / ROM non-volatile; cache faster than RAM, not a backup; CPU never touches secondary storage directly.
- ▶ Plotter, drum printer, sound card = output; scanner, light pen, digitizer = input; driver = software; BIOS boots.
- ▶ F7 checks spelling, F5 runs the show; Section Break changes headers; Mail Merge joins letters to a database.
- ▶ Formula starts with =; \$A\$1 is absolute; Select query retrieves; extension $\neq$ format.
- ▶ IPv6 = 128-bit; POP3 downloads, IMAP syncs; BCC hides recipients from each other; encryption $\neq$ signature.
- ▶ BGP is inter-domain; ICMP is diagnostic; HTTP/3 = QUIC over UDP; IT Act 2000, G2C/G2B/G2G/G2E.